INCH-POUND
MIL-PRF-22710/24D
21 October 2024
SUPERSEDING
MIL-PRF-22710/24C
20 November 2018

PERFORMANCE SPECIFICATION SHEET

SWITCHES, ROTARY (PRINTED CIRCUIT), (TOGGLE, BIDIRECTIONAL, SEALED, ILLUMINATED OR NONILLUMINATED), STYLE SRPC24

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this document and MIL-PRF-22710.

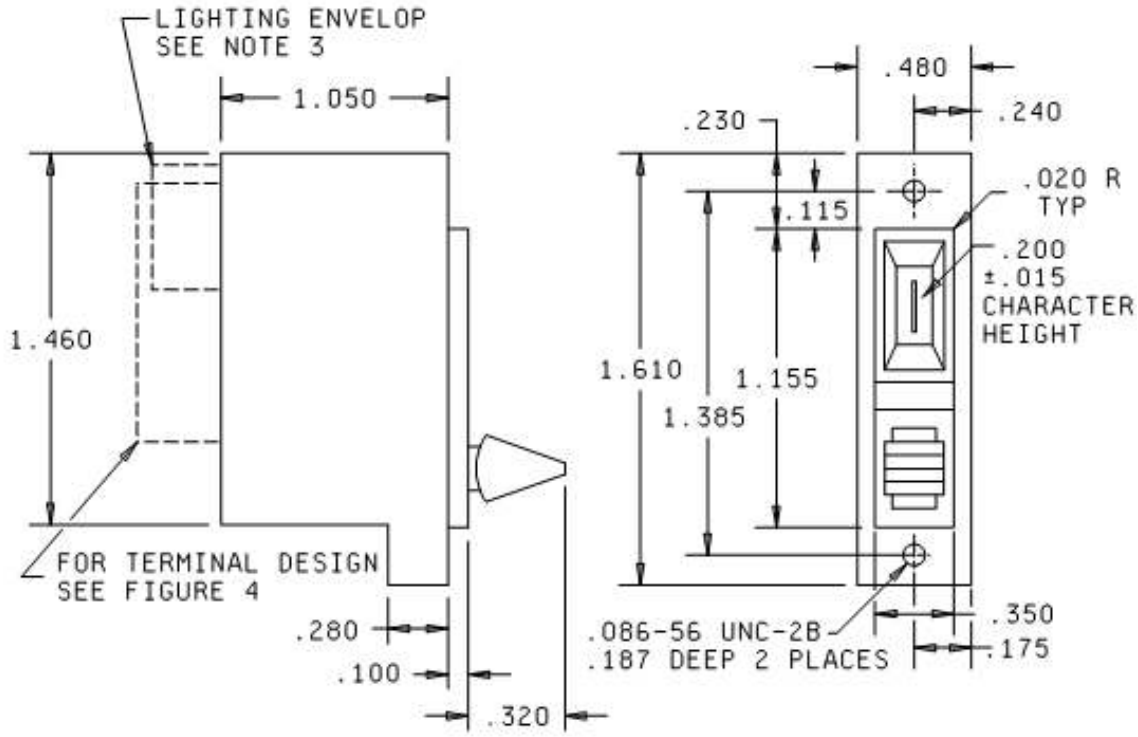
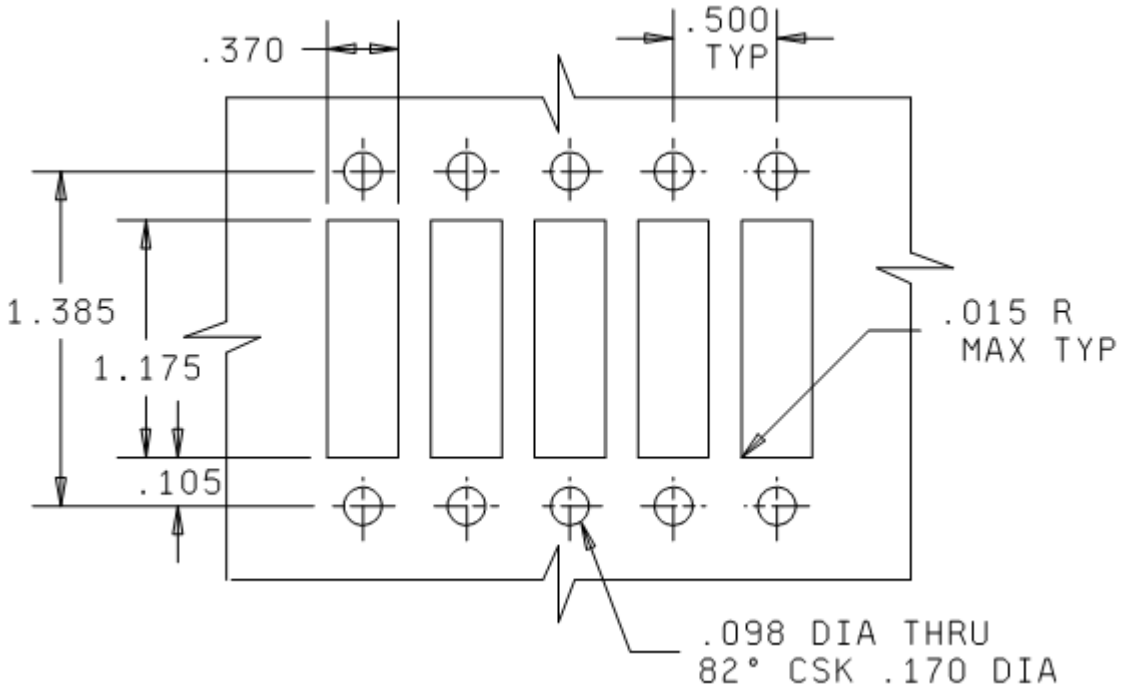


FIGURE 1. Style SRPC24 switch and panel cutout and installation data.

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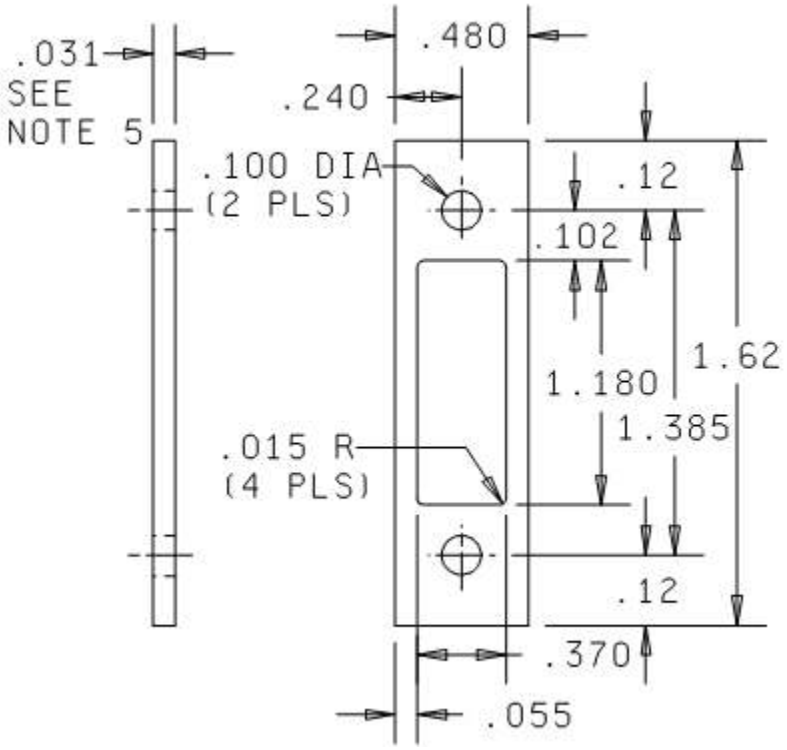
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NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .010.
3. Applicable to illuminated switches only. Allowance for lighting terminals for externally replaceable lamps, when internal lighting is specified.

FIGURE 1. Style SRPC24 switch and panel cutout and installation data - Continued.

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NOTES
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ± .005 for three place decimals and ±.02 for two place decimals.

FIGURE 2. Panel seal gasket.

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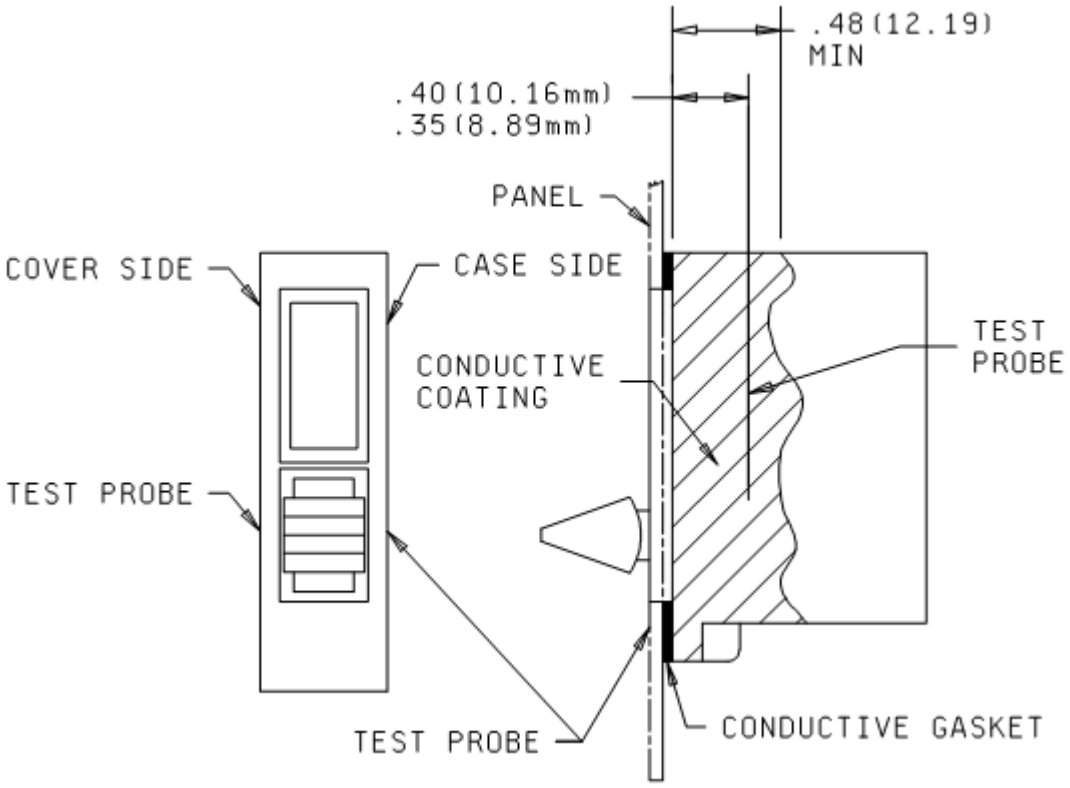


FIGURE 3. RFI shielded switch.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2 and 3.

Construction: D.

Angle of Throw: 36°

Temperature-life characteristic: B, except -55° low temperature for all switches and +71° C for lighted switches.

TABLE I. Electrical test load.

Test of condition	Resistive	
At atmospheric pressure	0.125 A	28 V dc and ac, 60 Hz
At reduced barometric pressure	0.125 A	28 V dc and ac, 60 Hz
Life (low level)	10 mA	30 mV

Vibration grade: 3 (15g, 10 - 2,000 Hz).

Shock types: M (100 g's) and H (high-impact). Methods I and II are applicable.

Altitude rating: C (70,000 feet).

Terminal design: See figure 3 and table III.

Contact resistance:
Initial: 50 milliohms, maximum.
Following life (rotational): 70 milliohms, maximum.

Luminace: Illuminated switches only.

Moisture resistance: MIL-STD-202-103, test condition A, with polarization voltage.

TABLE II. Photometric brightness.

Lamp	Switch filter	Brightness	Rating	
			Volts	Ampere
M6363/6 or equivalent	Clear	2.0 ± 1.0 foot-lamberts	5	0.115
	Red	1.0 ± 0.5 foot-lamberts	5	0.115
	Amber	2.0 ± 1.0 foot-lamberts	5	0.115

1/ MS6363/6 details are found in MIL-DTL-6363/6.

Operating force (tangential): 12 ounces minimum
36 ounces maximum

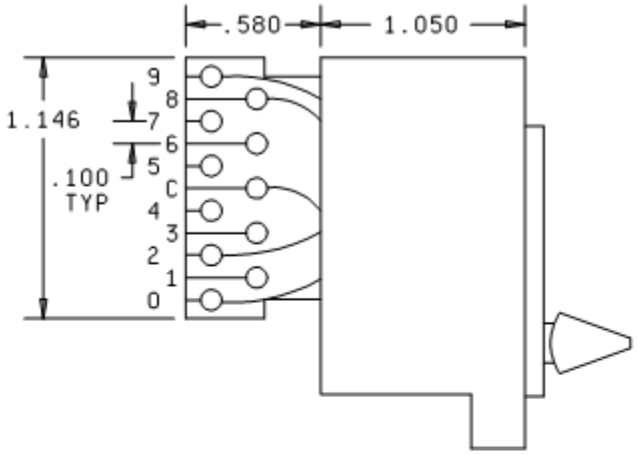
Seal: Method II, when specified in table I.

Dielectric withstanding voltage: 750 volts rms at atmospheric pressure
250 volts rms at reduced barometric pressure, test condition C.

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EMI/RFI shielding: Shielded switches (see table III) are intended for use in equipment required to meet the radiated (R) portions of MIL-STD-461. The conductive coating and conductive gasket shall be as shown in figure 3. When the switch is mounted facing outward to an electrically conductive panel, the resistance shall be measured between the RFI shielding on the switch and the mounting panel. The resistance shall not exceed 1 ohm.

Military part number: M22710/24- (dash number from table III).



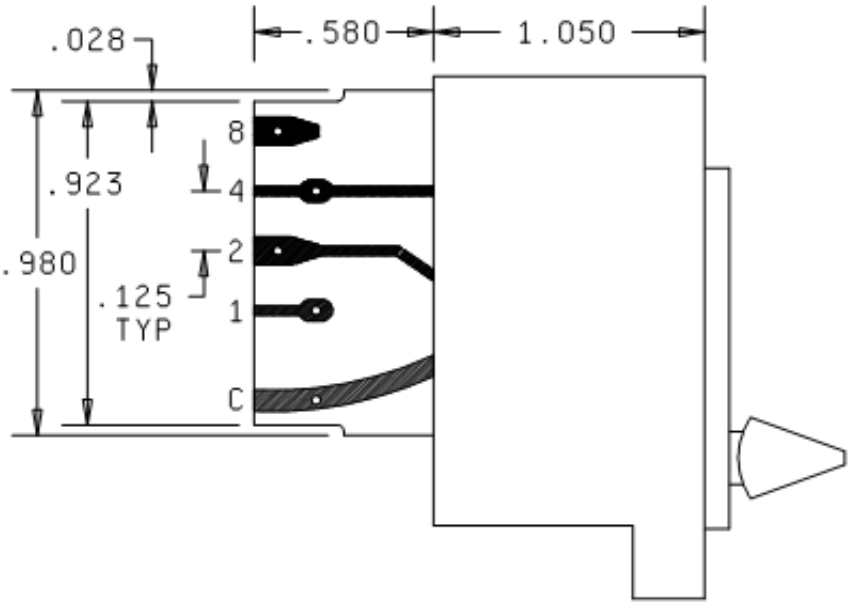
Truth table 1/ Straight decimal	
Dial Marking	Common c Conn to:
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9

1/ Code explanation: Dial reads 0 - Common C active with terminal 0 only.
Dial reads 1 - Common C active with terminal 1 only.
Dial reads 2 - Common C active with terminal 2 only.
and so forth.

Circuit configuration 1

FIGURE 4. Module and circuit configuration.

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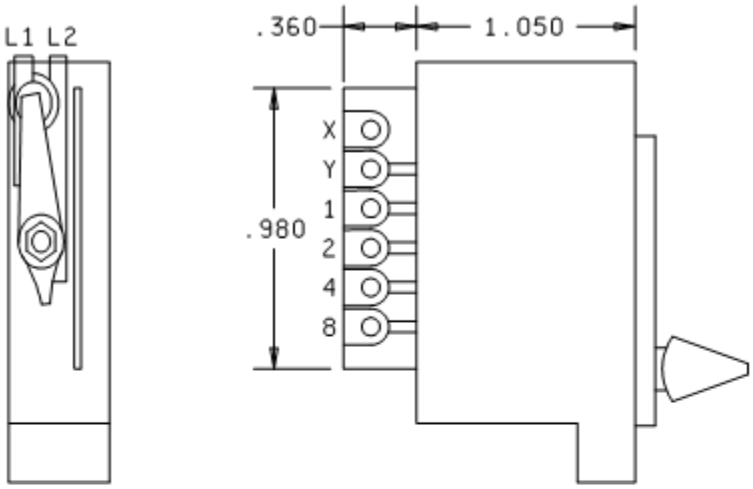
Truth table 1/ Binary coded decimal				
Dial marking	Common C Conn to:			
	1	2	4	8
0				
1	•			
2		•		
3	•	•		
4			•	
5	•		•	
6		•	•	
7	•	•	•	
8				•
9	•			•

1/ Code explanation: Dial reads 0 - All terminals non-active.
Dial reads 1 - Common C active with terminal 1 only.
Dial reads 2 - Common C active with terminal 2 only.
and so forth.

Circuit configuration 2

FIGURE 4. Module and circuit configuration - Continued.

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Truth table 1/ Binary coded octal with two commons				
Dial marking	Common X(•) Y(o) Conn to:			
	1	2	4	8
0	o	o	o	o
1	•	o	o	o
2	o	•	o	o
3	•	•	o	o
4	o	o	•	o
5	•	o	•	o
6	o	•	•	o
7	•	•	•	o
8	o	o	o	•
9	•	o	o	•

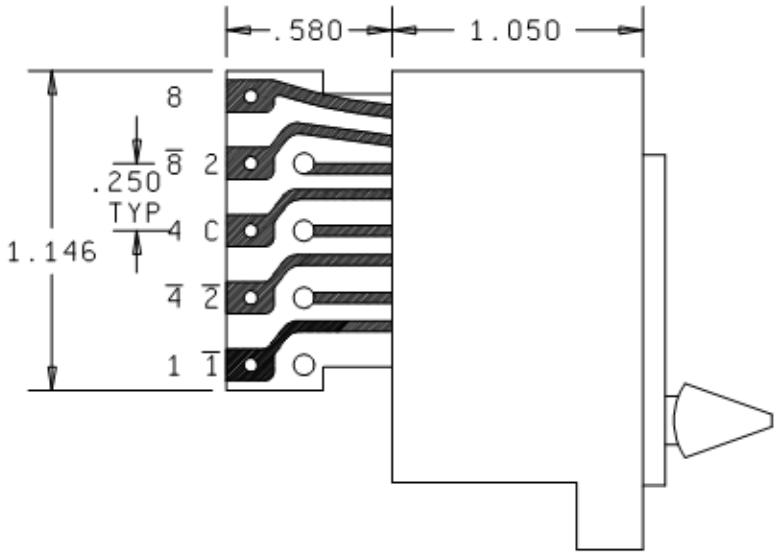
1/ Code explanation: Dial reads 0 - Common Y active with terminals 1, 2, 4 and 8 while common X is not connected to any terminals.
Dial reads 1 - Common X connected to terminal 1, while common Y is connected to terminals 2, 4 and 8.
Dial reads 2 - Common X connected to terminal 2, while common Y is connected to terminals 1, 4 and 8.
and so forth.

NOTES
1. Where lighting is specified, terminals shall be marked L1, L2.

Circuit configuration 3

FIGURE 4. Module and circuit configuration - Continued.

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Truth table 1/ Binary coded decimal plus compliment								
Dial marking	Common C Conn to:							
	1	2	4	8	$\overline{1}$	$\overline{2}$	$\overline{4}$	$\overline{8}$
0					•	•	•	•
1	•					•	•	•
2		•			•		•	•
3	•	•					•	•
4			•		•	•		•
5	•		•			•		•
6		•	•		•			•
7	•	•	•					•
8				•	•	•	•	
9	•			•		•	•	

1/ Code explanation: Dial reads 0 - Common C active with terminals 1, 2, 4 and 8
Dial reads 1 - Common C active with terminals 1, 2, 4 and 8.
Dial reads 2 - Common C active with terminals 2, 1, 4 and 8.
and so forth.

Circuit configuration 4

FIGURE 4. Module and circuit configuration - Continued.

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Table III. Part numbers and characteristics.

Part Number	Circuit configuration	Dial		illumination		Special features
M22710/24-		Background	Marking	Color	Location	
001	1	Black Opaque	White Opaque	---	---	---
002	1	Black Opaque	White Opaque	---	---	RFI Shielded
003	1	Black Opaque	White Translucent	Red	Back	---
004	1	Black Opaque	White Translucent	Red	Back	RFI Shielded
005	1	Black Opaque	White Translucent	Clear	Back	---
006	1	Black Opaque	White Translucent	Clear	Back	RFI Shielded
007	2	Black Opaque	White Opaque	---	---	---
008	2	Black Opaque	White Opaque	---	---	RFI Shielded
009	2	Black Opaque	White Translucent	Red	Back	---
010	2	Black Opaque	White Translucent	Red	Back	RFI Shielded
011	2	Black Opaque	White Translucent	Clear	Back	---
012	2	Black Opaque	White Translucent	Clear	Back	RFI Shielded
013	3	Black Opaque	White Opaque	---	---	---
014	3	Black Opaque	White Opaque	---	---	RFI Shielded
015	3	Black Opaque	White Translucent	Red	Back	---
016	3	Black Opaque	White Translucent	Red	Back	RFI Shielded
017	3	Black Opaque	White Translucent	Clear	Back	---
018	3	Black Opaque	White Translucent	Clear	Back	RFI Shielded
019	3	Black Opaque	White Translucent	Amber	Back	---
020	3	Black Opaque	White Translucent	Amber	Back	RFI Shielded
021	4	Black Opaque	White Opaque	---	---	---
022	4	Black Opaque	White Opaque	---	---	RFI Shielded
023	4	Black Opaque	White Translucent	Red	Back	---
024	4	Black Opaque	White Translucent	Red	Back	RFI Shielded
025	4	Black Opaque	White Translucent	Clear	Back	---
026	4	Black Opaque	White Translucent	Clear	Back	RFI Shielded

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Reference documents:

MIL-PRF-22710 MIL-STD-202-103
MIL-STD-461 MIL-DTL-6363/6

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Navy- EC
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DLA - CC

(Project 5930-2024-076)

Review activities:
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